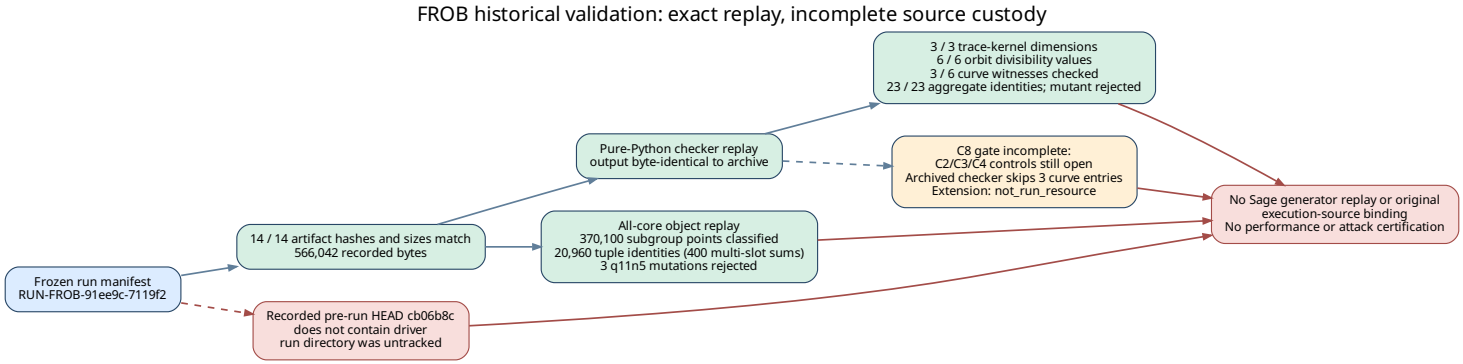
# Historical FROB checker replay and source custody

Validation audit, 22 October 2026

**Scope.** This checks frozen run `RUN-FROB-91ee9c-7119f2` at source tree `e83bb2f14414c9669360816aeee5876dfcfd4384`. It is an integrity audit, not a new FROB measurement or a research verdict. The run directory was copied before executing its pure-Python checker; frozen bytes were not edited.

| Check | Result | Limit |
|---|---|---|
| Archived artifact hashes and sizes | 14/14 match; 566,042 bytes | Storage integrity only |
| Checker output | Byte-identical replay | Same archived checker source |
| Trace-kernel dimension; orbit divisibility | 3/3; 6/6 | Recorded core cells |
| Curve point/order/Frobenius witnesses | 3/6 checked | Other three skipped for size |
| Aggregate rational identities | 23/23; mutant rejected | Counts treated as inputs |
| New all-core object-arm replay | 370,100 points; 20,960 tuples | 400 multi-slot sums; 3 q11 mutants |
| Remaining C8 work | Open | C2/C3/C4 controls |
| Extension cell | `not_run_resource` | 540-second historical checkpoint |




**Source custody.** The manifest records pre-run HEAD `cb06b8c20c2d6559041c45c4a4b29dd7f132fdf5` and says the run directory was untracked at execution start. The driver is absent from that commit. Current-schema `code.commit`, `code.command`, and `environment` fields are absent, although separate command and environment artifacts are hash-pinned. Their presence does not bind the executed driver to the recorded HEAD.

**Boundary and next check.** The archived checker's `tuple_sum_reverifications` label overstates its code: its three `ok` entries check curve witnesses, not tuple sums or control arms. The new core replay verifies all three object arms with the archived checker's field and curve primitives; the controls remain open. Thus C8 and the success gate are unestablished. Neither replay binds the original Sage execution source or proves attack performance. Recover a contemporaneous source snapshot or run a fresh source-bound replay under the frozen controls. The Coordinator must adjudicate the original record without assigning the pre-run HEAD as a false execution-source commit. Receipts: `audit.json`, `core-object-replay.json`.